

Coursework Title: Design and implement an image manipulation and communication system

Module Name: Embedded Systems
Module Code: 7430MESI
Level: 7
Credit Rating: 20
Weighting: 30%
Lecturer: Sami Siddiqui
Contact: If you have any issues with this coursework you may contact your lecturer.
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Issue Date: 27/2/26
Hand-in Date: Demonstration and Viva on 21st April '26;
Report on 3rd May '26
Hand-in Method: via CANVAS & Lab Demo
Feedback date: Wk 33
Feedback Method: Oral and written
Programmes: MSc Embedded Systems and IC design/MEng Electronic Engineering
Learning Outcomes:

- 1 Design and implement complex integrated hardware and software solutions to engineering problems

The Task.

Using the appropriate Arduino based circuits (UNO, OLED, Bluetooth) you are asked to design and implement a system with the following objectives:

1. Find or generate a simple asymmetrical image (e.g. the letter F, G, J, K, L, P, Q, R) which can be displayed on the OLED.
2. Load the image into the Arduino.
3. Display the image on the OLED.
4. Manipulate the image under the control of an external input (e.g. switches, joystick etc). Manipulation includes, moving, rotating, or transposing the image.
5. Send the image to another similar device.
6. Receive an image from another similar device.
7. Store the received image.
8. Display the received image on the OLED.
9. Manipulate the received image under the control of an external input.
10. Measure or estimate the system's power consumption in various operating modes (idle, image manipulation, Bluetooth transmission).

NOTE: There are apps available which can act as the second device to demonstrate sending and receiving of the image.

Marking Scheme: Demo and Viva (10% total)

Demonstration of working system	33
Explanation of system	33
Answers to design and results-related questions	33

Marking Scheme: Report Marking scheme: (90% total)

Introduction, Introduce the topic and report layout	10
Background and Method employed. Discussion of technologies, deployment, use of hardware and software (e.g., Arduino UNO, OLED, Bluetooth), parameters. Arduino Platform, hardware accessories, language used, development environment, etc.	20
Design Discussion of the software design. Use of flowcharts. Explanation of code snippets, clarification of which code has been written by the author, and which has been reused from other sources. Discussion of hardware design. Circuit diagrams, design decisions, choice of components, etc.	15
Environmental Sustainability Assess the system's environmental impact across its life cycle: component sourcing, manufacturing, use-phase, and end-of-life disposal. Evaluate the system's energy efficiency based on measurements or estimates. Describe design strategies used to reduce power dissipation and lower the carbon footprint	10
Social and Ethical Responsibility Recommend safe, responsible recycling or disposal of obsolete hardware. Discuss how your project contributes to sustainable development goals. Consider ethical aspects, such as equitable access to technology.	10
Discussion of the tests Description of tests executed. Discussion of test results from executed tests. Testing rationale including: why were these executed? Other tests which were considered and not executed.	15
Conclusion on performance What do your results show about the performance of your system? Which aspects of your project worked well, and which could have been done differently.	5
Code Listing with appropriate comments. <u>Your code listing should be included as text (not screen shots). Submissions which do not conform to this will be rejected.</u>	10

NOTES:

1. Using material from other sources.

Arduino is a very widely used product set. As such, there are available several sources of hardware design, software design, and code. In your project work, you may decide to reuse elements from other sources as a basis for, or to compliment your own work. In that case, you should CLEARLY indicate which work has come from other sources and where your contribution is. In addition to showing this clear distinction, the external work should be appropriately referenced.

Your code listing should be included as text (not screen shots). Submissions which do not conform to this will be rejected.

Guide to Performance Criteria (The Module Leader is advised to delete sections not applicable to the coursework set and if necessary modify the criteria accordingly)

70% and above:

Your work must be of outstanding quality and fully meet the requirements of the coursework specification and learning outcomes stated. You must show independent thinking and apply this to your work showing originality and consideration of key issues. There must be evidence of wider reading on the subject.

Key words which may describe a coursework at this level include: *appraises, compares, concludes, contrasts, criticizes, critiques, defends, discriminates, evaluates, explains, interprets, justifies, relates, supports.*

60% - 70%:

Your work must be of good quality and meet the requirements of the coursework specification and learning outcomes stated. You must demonstrate some originality in your work and show this by applying new learning to the key issues of the coursework. There must be evidence of wider reading on the subject.

Key words which may describe a coursework at this level include: *categorizes, combines, compiles, creates, devises, generates, modifies, reconstructs, identifies, illustrates, outlines, synthesizes.*

50% - 60%:

Your work must be comprehensive and meet all of the requirements stated by the coursework specification and learning outcomes. You must show a good understanding of the key concepts and be able to apply them to solve the problem set by the coursework. There must be enough depth to your work to provide evidence of wider reading.

Key words which may describe a coursework at this level include: *demonstrates, changes, applies, operates, produces, predicts, shows, solves, uses, translates, comprehends, converts, generalizes.*

40% - 50%:

Your work must be of a standard that meets the requirements stated by the coursework specification and learning outcomes. You must show a reasonable level of understanding of the key concepts and principles and you must have applied this knowledge to the coursework problem. There should be some evidence of wider reading.

Key words which may describe a coursework at this level include: *comprehends, defines, describes, identifies, knows, labels, lists, matches, outlines, recalls, recognizes, reproduces, selects, states, rewrites.*

Below 40%:

Your work is of poor quality and does not meet the requirements stated by the coursework specification and learning outcomes. There is a lack of understanding of key concepts and knowledge and no evidence of wider reading.

Personal (Extenuating) Circumstances

If something serious happens that means that you will not be able to complete this assignment, you need to contact the module leader as soon as possible. There are a number of things that can be done to help, such as extensions, waivers and alternative assessments, but we can only arrange this if you tell us. To ensure that the system is not abused, you will need to provide some evidence of the problem. More guidance is available at

<https://policies.ljmu.ac.uk/UserHome/Policies/PolicyDisplay.aspx?&id=254&l=1>.

Any coursework submitted late without the prior agreement of the module leader will receive 0 marks.

Academic Misconduct

The University defines Academic Misconduct as 'any case of deliberate, premeditated cheating, collusion, plagiarism or falsification of information, in an attempt to deceive and gain an unfair advantage in assessment'. The School takes Academic Misconduct very seriously and any suspected cases will be investigated through the University's standard policy

(<https://policies.ljmu.ac.uk/UserHome/Policies/PolicyDisplay.aspx?&id=252&l=1>). If you are found guilty, you may be expelled from the University with no award.

It is your responsibility to ensure that you understand what constitutes Academic Misconduct and to ensure that you do not break the rules. If you are unclear about what is required, please ask.

Cheating includes:

- (i) any form of communication with, or copying from, any other source during an examination;
- (ii) communicating during an examination with any person other than an authorised member of staff;
- (iii) introducing any written, printed or other material into an examination (including electronically stored information) other than that specified in the rubric of the examination paper;
- (iv) gaining access to unauthorised material in any way during or before an assessment;

- (v) the use of mobile phones or any other communication device during an assessment or examination;
- (vi) the submission of false claims of previously gained qualifications, research or experience in order to gain credit for prior learning;
- (vii) the falsification of research data, the presentation of another's data as one's own, and any other forms of misrepresentation in order to gain advantage;
- (viii) the submission of work for assessment that has already been submitted as all or part of the assessment for another module without the prior knowledge and consent of the Module Leader for the subsequent assessments;
- (ix) the submission of material purchased or commissioned from a third party, such as an essay-writing service, as one's own.

Plagiarism is defined as the representation of the work, artefacts or designs, written or otherwise, of any other person, from any source whatsoever, as the student's own. Examples of plagiarism may be as follows:

- i) the verbatim copying of another's work without clear identification and acknowledgement including the downloading of materials from the Internet without proper referencing of materials;
- ii) the paraphrasing of another's work by simply changing a few words or altering the order of presentation, without clear identification and acknowledgement;
- iii) the unidentified and unacknowledged quotation of phrases from another's work;
- iv) the deliberate and detailed presentation of another's concept as one's own.

Collusion includes:

- (i) the conscious collaboration, without official approval, between two or more students in the preparation and production of work which is ultimately submitted by each in an identical or substantially similar form and/or is represented by each to be the product of his or her individual efforts;
- (ii) collusion also occurs where there is unauthorised co-operation between a student and another person in the preparation and production of work which is presented as the student's own.

For more information you are directed to following the University web pages:

- Information regarding **academic misconduct**:
<https://www.ljmu.ac.uk/about-us/public-information/student-regulations/academic-misconduct>
- Information on **study skills**:
<https://www.ljmu.ac.uk/microsites/library/skills-ljmu>
- Information regarding **referencing**:
<https://www.ljmu.ac.uk/microsites/library/subject-resources/referencing>